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SKG 12 Mechanical Engineering

Predictive Maintenance Technique – Part 1 (Failure Mechanism), L2

Time to Failure and Probability of Failure

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Time to Failure and Probability of Failure

Learning Objectives

By the end of this session, you should be able to:

- Be familiar with basic reliability concepts
- Understand the traditional approach on Reliability prediction i.e. probabilistic approach

Objectives of Reliability and Maintainability Engineering

- To study, characterize, measure and analyze the [failure](#) and [repair](#) of systems.
- Aims to improve operational use of systems and equipment by increasing design life, [eliminating or reducing the likelihood of failures and safety risks](#), and reducing downtime, thereby increasing available operating time.
- Closely related to R&M is [Availability](#), which measures the combined effect of BOTH failure and the repair process.
- So, in terms of performance a production system or an equipment has three important characteristics:
 - Reliability,
 - Maintainability, and
 - Availability.

Reliability

Definition:

Probability of an item to perform a required *function* under given conditions for a given **time interval**¹.

¹ISO 20815, first edition, 2008, page 2.

- Key words:
 - probability,
 - an item,
 - to perform a required *function* (**any function**),
 - given conditions ($p, T, \dot{Q}, h, \%, F, \dot{\omega}, V, I, P$), and
 - for a given **time interval**.



Reliability

- Reliability is, in simple terms, the probability of success.
- Longer mission time results in lesser Reliability $\rightarrow 0$.
- Mathematical expression of Reliability:

$$\text{Reliability} = R(t) = 1 - F(t) = 1 - \int_0^t f(t) dt$$

Where:

$R(t)$ is Reliability at time “ t ”,

$F(t)$ is Probability of Failure at time “ t ” – i.e. the Cumulative Distribution Function (CDF),

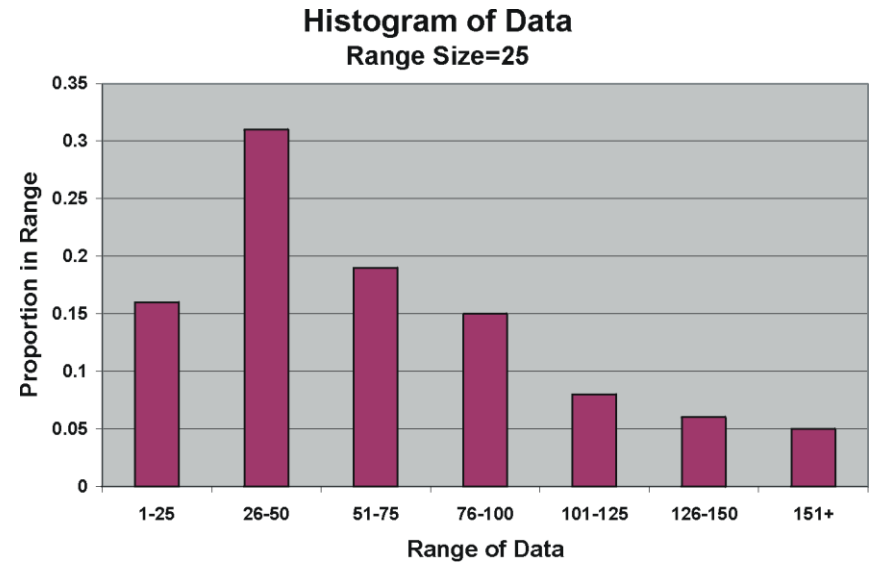
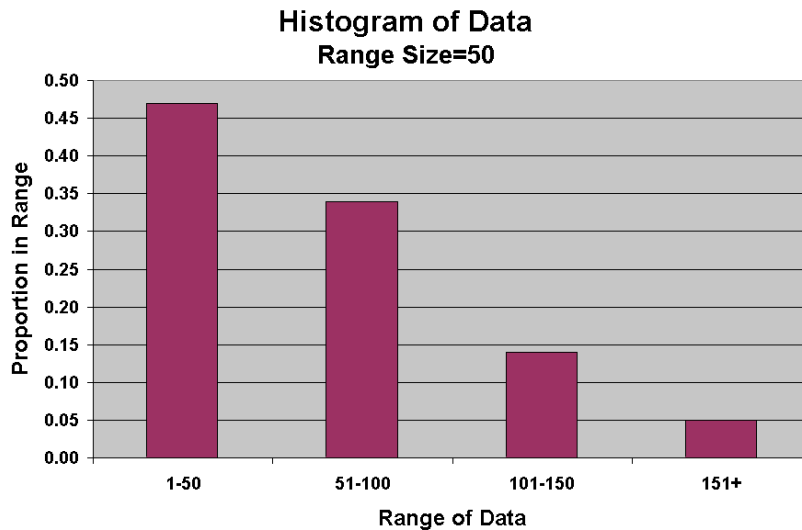
$f(t)$ is the Probability Density Function (PDF) at time “ t ” – determined by the Time(s)-to-Failure (TTF).

Reliability – Probability Density Function, $f(t)$

We use an arbitrary distribution of failure times to explain the PDF...

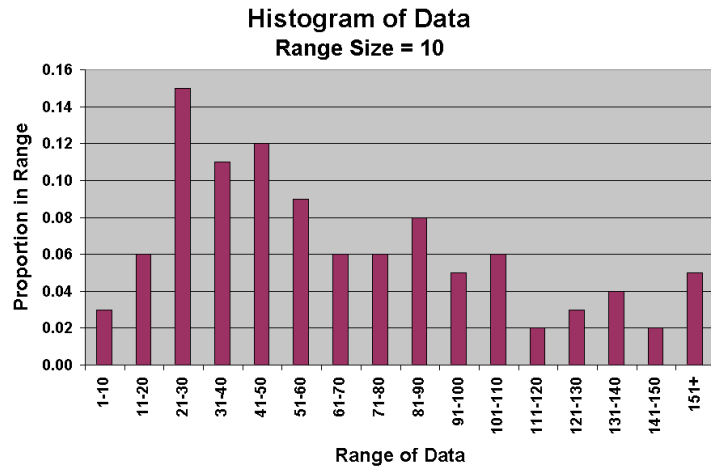
We are familiar with histograms:

The same dataset can also be represented by another histogram:

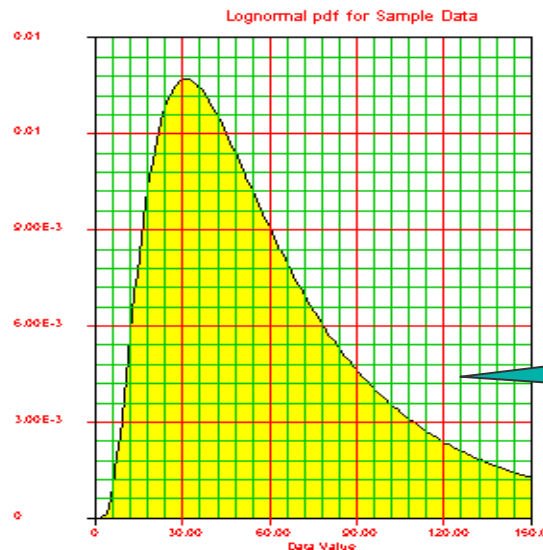


Reliability – Probability Density Function, $f(t)$

And yet, another histogram:



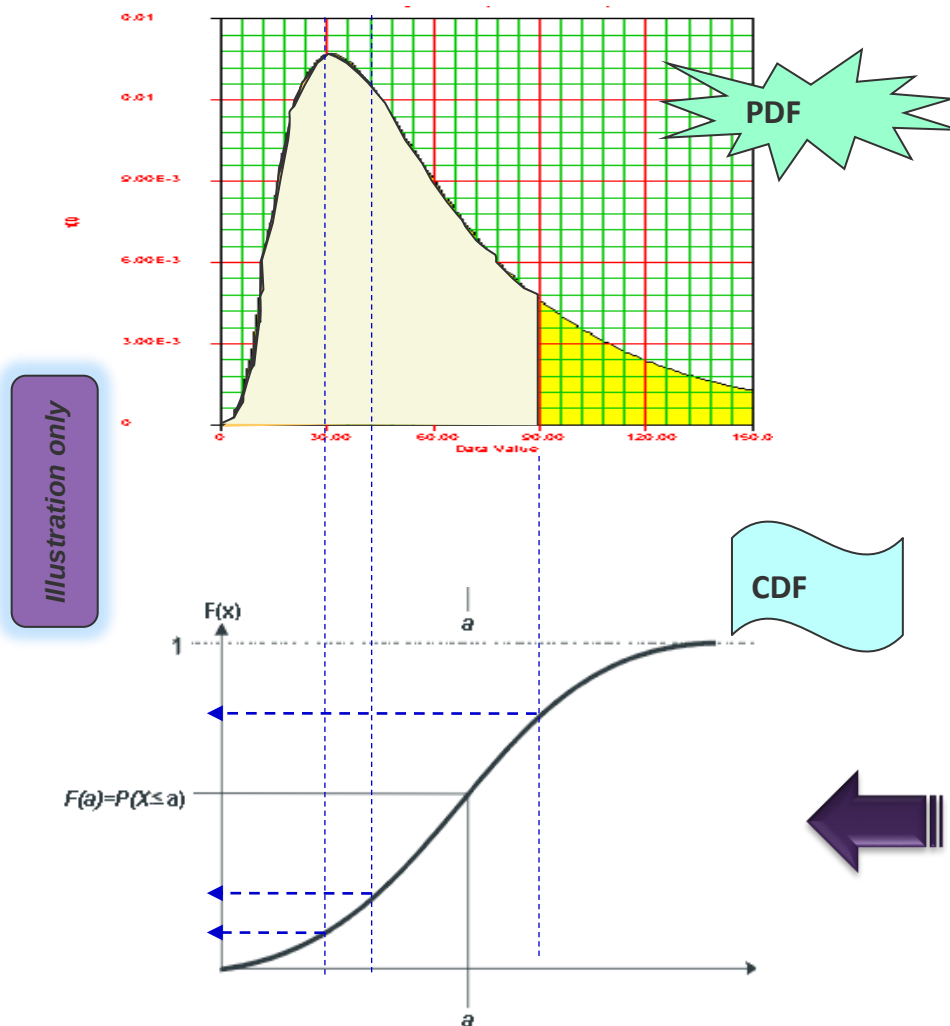
What happens if we were to plot the histogram in continuous form?



- The plot is called the **Probability Density Function**
- Using “Proportion in Range” as the Y-axis, the values of Y for each corresponding failure times (X-axis) would add up to 1.

The PDF gives us a visual representation of the failure distribution

Reliability – Cumulative Distribution Function (CDF), $F(t)$



- The fact that all Y-values add up to 1 gives us the following property of the PDF:

$$[\text{Area under the PDF}] = \int_{t=0}^{\infty} f(t)dt = 1$$

- If we take any instantaneous time, t_1 , we can calculate the area under the PDF curve between $t=0$ to $t=t_1$

$$[\text{Area between } t = 0 \text{ to } t = t_1] = \int_{t=0}^{t_1} f(t)dt$$

- If we then plot each area against its corresponding time, $t_1, t_2, \dots, t_{\infty}$, we would get a graph shown on the left.

- This graph is called ...

The Cumulative Distribution Function (CDF)

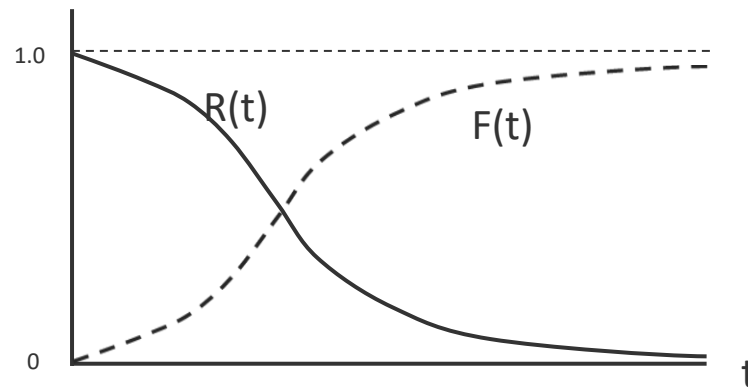
- The CDF, $F(t)$ is also known as...the **Failure Function**

Reliability

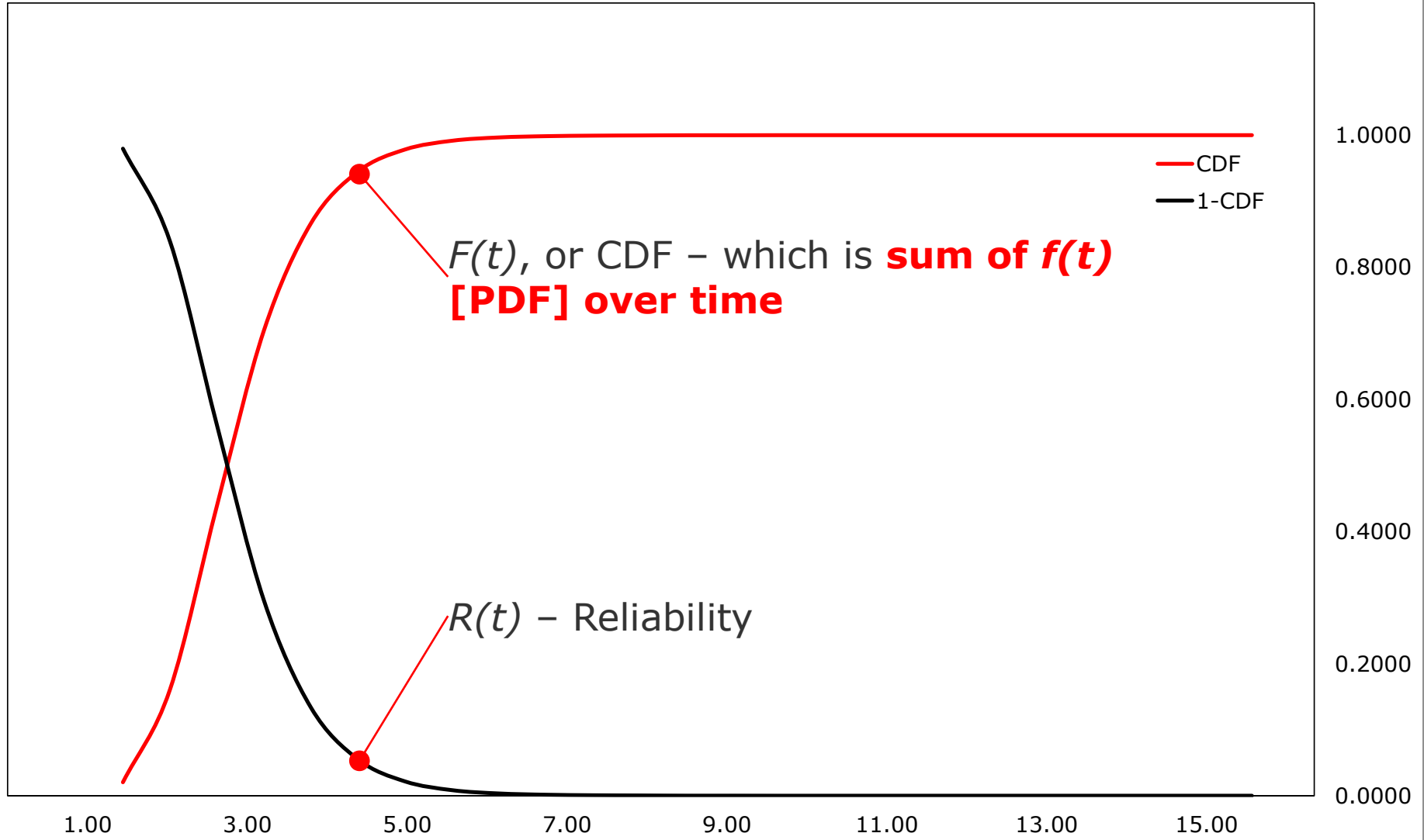
- Recall that...

$$R(t) = 1 - F(t) = 1 - \int_{t=0}^t f(t)dt = \int_t^{\infty} f(t)dt$$

- Once we know the CDF, $F(t)$, finding the Reliability function, $R(t)$ is just a matter of applying one of probability rules.
- Graphically, an example of $R(t)$...



Reliability



Maintainability

Definition:

Probability of an item under given conditions of use, to be retained in, or restored to, a state in which it can perform a required *function*, when maintenance is performed under given conditions and using stated procedures and resources¹.

¹IEC 60300-2, second edition, 2004, page 31.

- Key words:
 - probability,
 - an item,
 - given conditions of use (resulting in extent of damage),
 - retained in, or restored to [**within a specified time**], a state (As Good As New, As Bad As Old, or Minimal Repair),
 - can perform a required *function*,
 - maintenance is performed under given conditions (e.g. lighting, access),
 - stated procedures, and
 - resources (skilled or apprentice technician, spare parts, and use of special tools).

Maintainability

- Maintainability is, in simple terms, the probability of success.
- Shorter target for repair/restoration time results in lesser Maintainability $\rightarrow 0$.
- Mathematical expression of Maintainability:

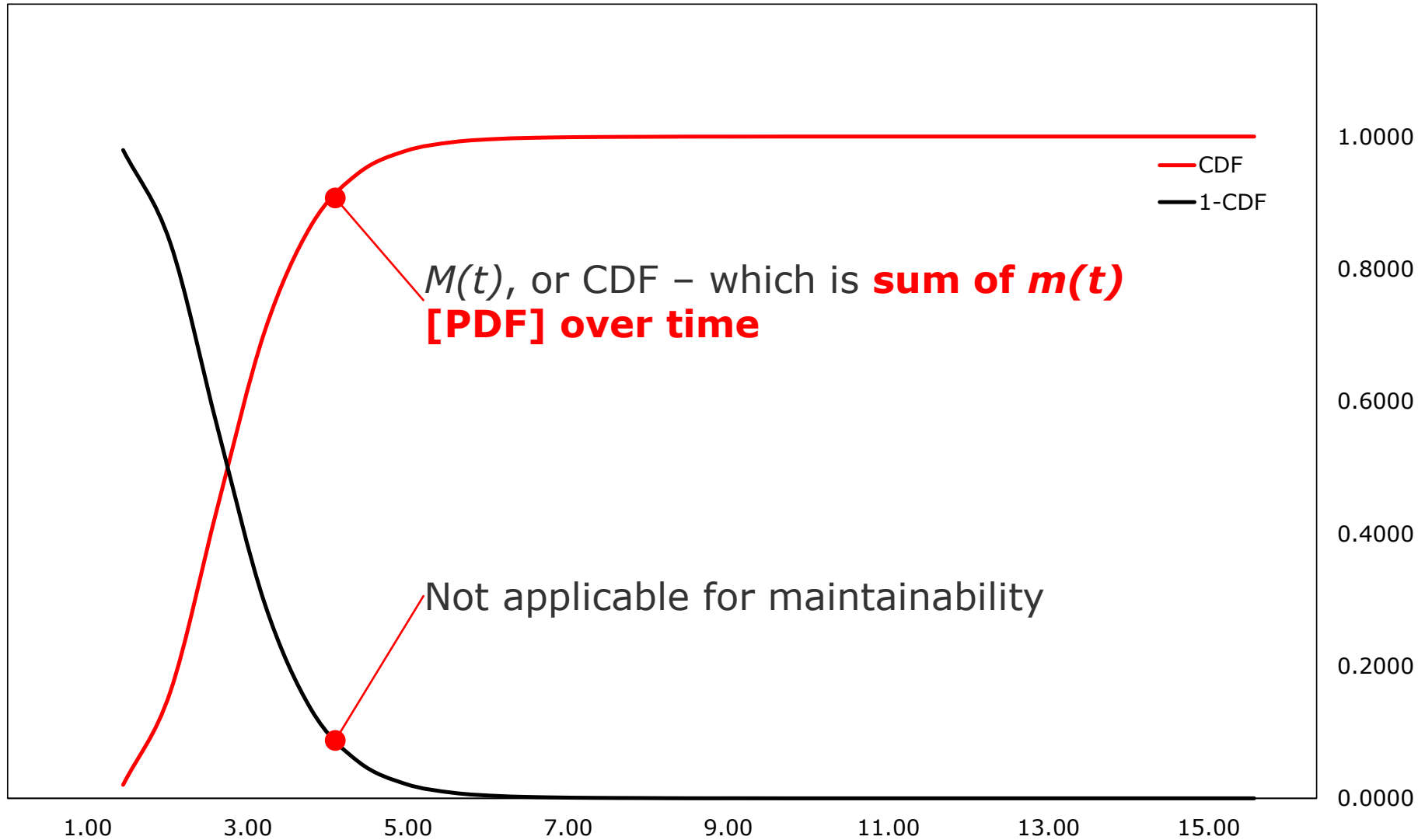
$$\text{Maintainability} = M(t) = \int_0^t m(t) dt$$

Where:

$M(t)$ is Maintainability at time “ t ”, i.e. the CDF,

$m(t)$ is the PDF at time “ t ” – determined by the Time(s)-to-Repair (TTR), or Time(s)-to-Restoration (TR).

Maintainability



Maintainability

ISO 14224:2006(E)

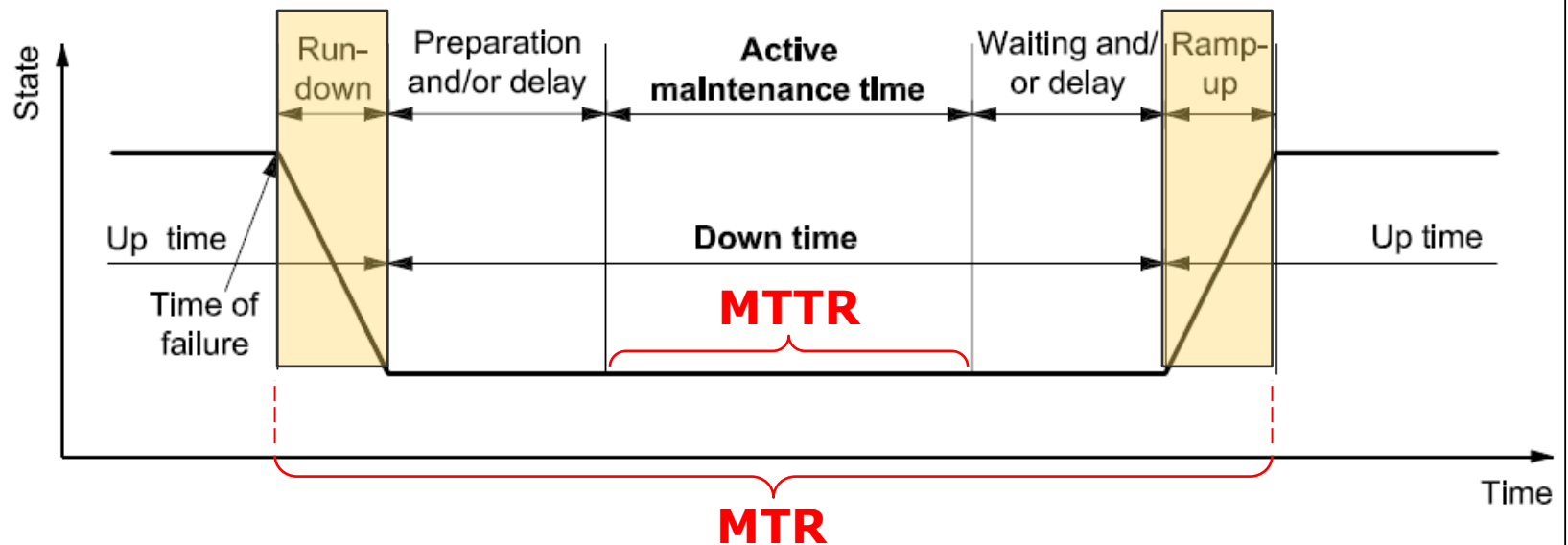


Figure 4 — Maintenance times

Depending on situation, "Run-down" and "Ramp-up" may be included in MTR calculation

Availability

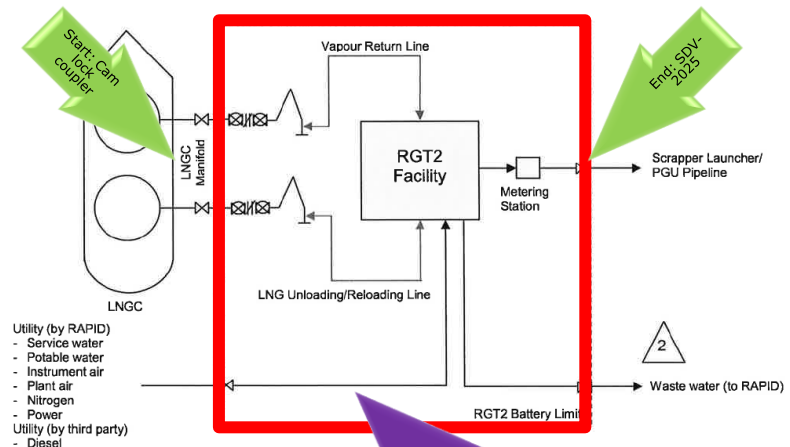
Definition:

Probability of an item to be in a state to perform a required function under given conditions at a given **instant of time**, or in average over a given time interval, assuming that the required external¹ resources are provided².

¹External to the boundary of study or interest. If supply availability is of concern and included in the study, then it is no longer considered "external".

²ISO 20815, first edition, 2008, page 2.

- Key words:
 - probability,
 - an item,
 - be(ing) in a state,
 - perform(ing) a required *function*,
 - given conditions (p , T , $\dot{Q}$, h , $\%$, F , $\dot{\omega}$, V , I , P),
 - at a given **instant of time**
 - **or** in average over a given time interval, and
 - external resources are provided.

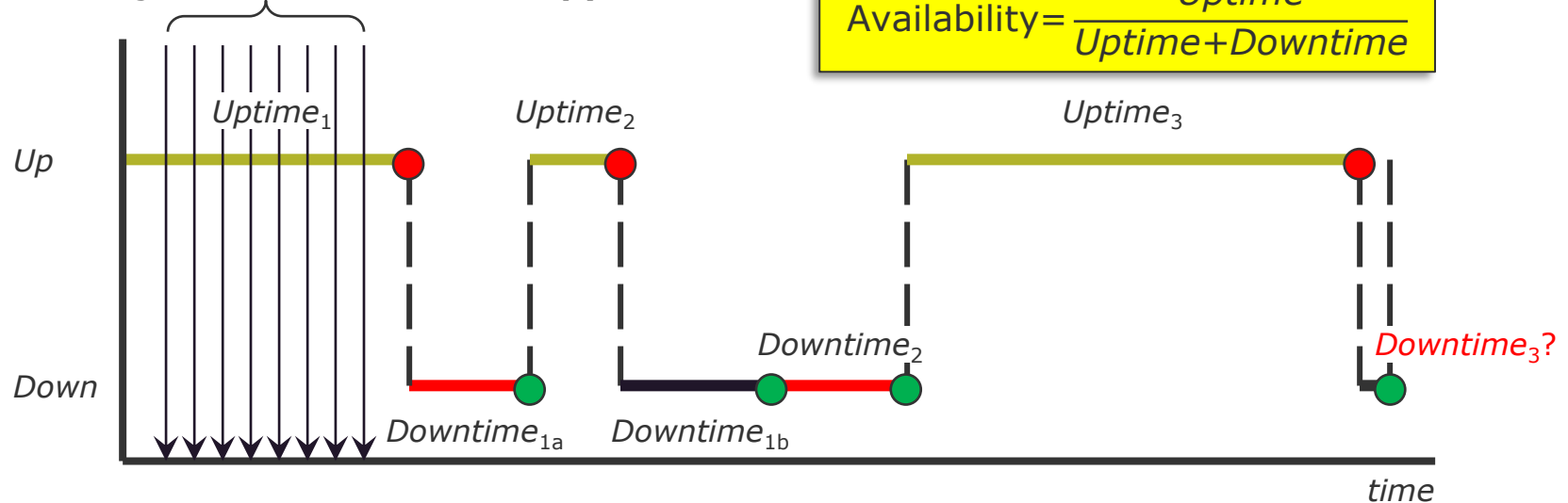


Studied system are:

- RGT2 Facility,
- Loading Arms,
- Metering Station,
- Internal Power Distribution.

Availability

"... at a given instant of time ..." – $A(t)$



- Uptime (operating, i.e. *functioning*). Excludes, unplanned and planned stoppages (i.e. run-time). Uptime is a simplified measure of Reliability.
 - Downtime (planned maintenance).
 - Downtime (unscheduled).
 - Downtime (external – to the equipment – reasons, e.g. put on standby, no demand).
- } Includes the duration of, planned and unplanned downtimes. These are simplified measure of Maintainability
- | Run-down or ramp-up (terminology from ISO 14224:2006, page 21).
- "Switch ON" (terminology from ReliaSoft®), i.e. demand imposed.
- "Switch OFF" (terminology from ReliaSoft®), i.e. demand removed.

Availability

$$\text{Availability} = \frac{\text{Uptime}}{\text{Uptime} + \text{Downtime}}$$

- “Uptime” is a measure of Reliability (Mean Time Between Failure - MTBF)
- “Downtime” is a measure of Maintainability (Mean To Restore - MTR)
- You can have:
 - high Reliability and poor Maintainability equipment that has same Availability as,
 - poor Reliability and high Maintainability equipment.

$$A = \frac{MTBF}{MTBF + MTR}$$





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Thank you